

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location CyrusOne Austin Data Center II, TX -- station KAUS, America/Chicago
Committed pair OSM 455171103 -> 379204766
 CyrusOne Austin Data Center III / CyrusOne Austin Data Center II
Facade gap 149.9 m between the two halls
Weather record 43,746 real hourly records from KAUS
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise
 0.4627 C at 285 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-10-29 (knife_edge)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 24 hours with 1 mode change. The reactive incumbent took 8 hours -- 3 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
 the incumbent broke its own switch budget 1 time(s) to stay safe
 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	15.560	18.0	15.944	1.048
01	mechanical	yes	switch budget	15.275	18.0	14.587	1.201
02	mechanical	yes	switch budget	15.275	18.0	14.823	1.036
03	mechanical	yes	switch budget	14.737	18.0	13.713	0.894
04	mechanical	yes	switch budget	14.445	18.0	14.311	0.754
05	mechanical	yes	switch budget	14.720	18.0	14.747	0.668
06	mechanical	yes	switch budget	14.557	18.0	14.747	0.846
07	mechanical	yes	switch budget	15.280	18.0	14.600	0.883
08	mechanical	no	dry-bulb	18.055	18.0	15.708	1.681
09	mechanical	no	dry-bulb	20.912	18.0	17.276	2.623

10	mechanical	no	dry-bulb	22.221	18.0	18.386	2.978
11	mechanical	no	dry-bulb	22.514	18.0	20.601	1.910
12	mechanical	no	dry-bulb	22.780	18.0	21.681	1.173
13	mechanical	no	dry-bulb	23.476	18.0	22.231	1.058
14	mechanical	no	dry-bulb	24.579	18.0	23.330	0.919
15	mechanical	no	dry-bulb	24.170	18.0	23.378	0.852
16	mechanical	no	dry-bulb	23.889	18.0	23.463	0.858
17	mechanical	no	dry-bulb	23.793	18.0	22.518	1.117
18	mechanical	no	dry-bulb	21.386	18.0	19.513	1.379
19	FREE	yes	-	17.440	18.0	12.495	1.530
20	FREE	yes	-	15.751	18.0	10.196	1.732
21	FREE	yes	-	14.995	18.0	10.620	1.747
22	FREE	yes	-	10.702	18.0	8.922	1.565
23	FREE	yes	-	10.268	18.0	10.117	1.210

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.560 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.275 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.275 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.737 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.445 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.720 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.557 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.280 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.055 C against a 18.0 C limit, so it fails by 0.055 C. It would take a limit 0.055 C higher, or a bound 0.055 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.912 C against a 18.0 C limit, so it fails by 2.912 C. It would take a limit 2.912 C higher, or a bound 2.912 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.221 C against a 18.0 C limit, so it fails by 4.221 C. It would take a limit 4.221 C higher, or a bound 4.221 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.514 C against a 18.0 C limit, so it fails by 4.514 C. It would take a limit 4.514 C higher, or a bound 4.514 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.780 C against a 18.0 C limit, so it fails by 4.780 C. It would take a limit 4.780 C higher, or a bound 4.780 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.476 C against a 18.0 C limit, so it fails by 5.476 C. It would take a limit 5.476 C higher, or a bound 5.476 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.579 C against a 18.0 C limit, so it fails by 6.579 C. It would take a limit 6.579 C higher, or a bound 6.579 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.170 C against a 18.0 C limit, so it fails by 6.170 C. It would take a limit 6.170 C higher, or a bound 6.170 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.889 C against a 18.0 C limit, so it fails by 5.889 C. It would take a limit 5.889 C higher, or a bound 5.889 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.793 C against a 18.0 C limit, so it fails by 5.793 C. It would take a limit 5.793 C higher, or a bound 5.793 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.386 C against a 18.0 C limit, so it fails by 3.386 C. It would take a limit 3.386 C higher, or a bound 3.386 C tighter, to change this hour.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.440 C, which is 0.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.529 C, and the margin is measured, not chosen: 1.529 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.751 C, which is 2.249 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.732 C, and the margin is measured, not chosen: 1.732 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.995 C, which is 3.005 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.747 C, and the margin is measured, not chosen: 1.747 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.702 C, which is 7.298 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.565 C, and the margin is measured, not chosen: 1.565 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.268 C, which is 7.732 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.210 C, and the margin is measured, not chosen: 1.210 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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